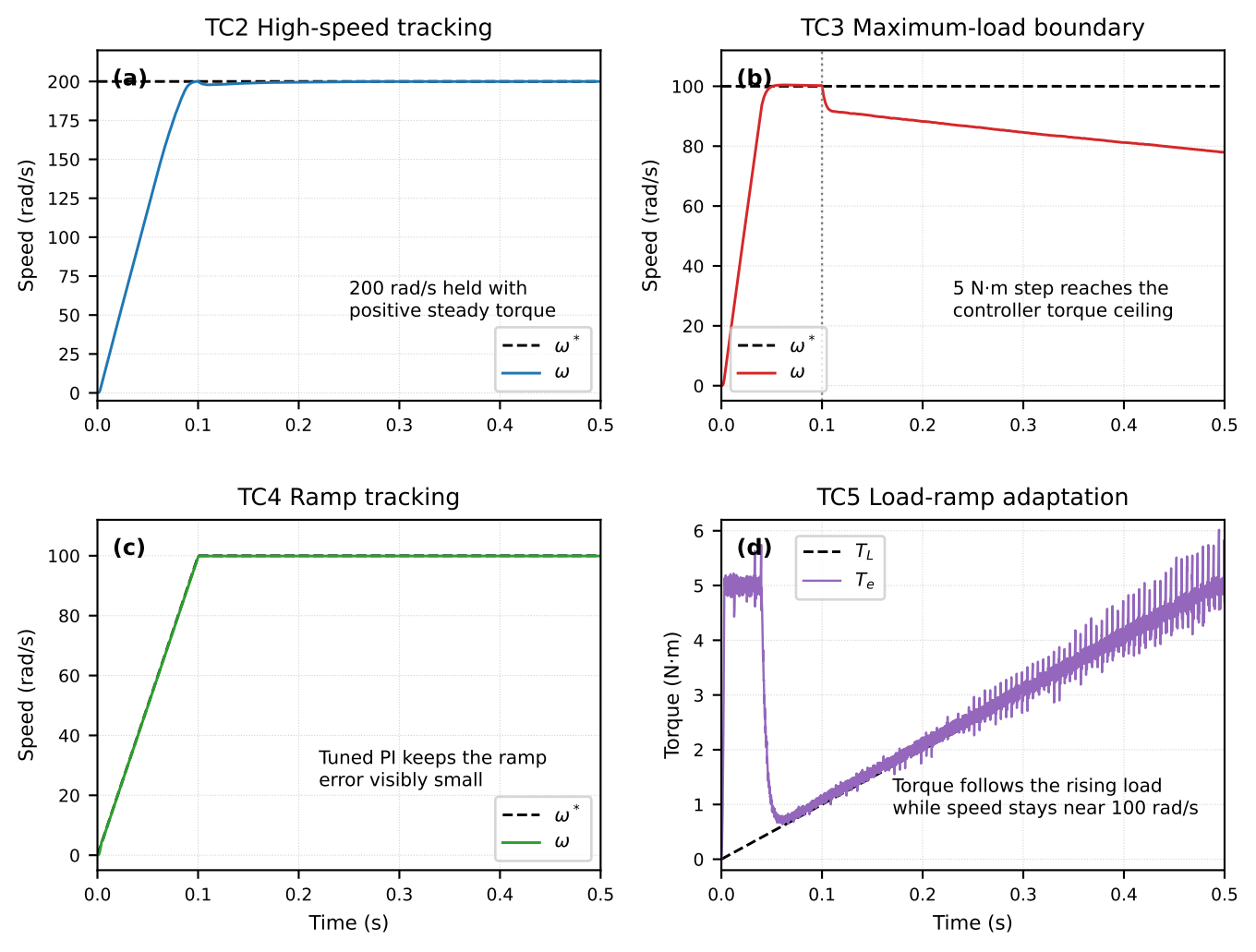

TC2 High-speed tracking
(a)
Speed (rad/s)
200 rad/s held with
positive steady torque
ω^*
ω
TC3 Maximum-load boundary
(b)
Speed (rad/s)
5 N·m step reaches the
controller torque ceiling
ω^*
ω
TC4 Ramp tracking
(c)
Speed (rad/s)
Tuned PI keeps the ramp
error visibly small
ω^*
ω
Time (s)
TC5 Load-ramp adaptation
(d)
T_L
T_e
Torque (N·m)
Torque follows the rising load
while speed stays near 100 rad/s
Time (s)